

Model validation and 1D models

Modelling 2025

Class notes for 30 October 2025

See also: [Thackner et al. 2024](#).

Concepts of Model Verification and Validation

Ben H. Thacker*

Scott W. Doebling

Francois M. Hemez

Mark C. Anderson

Jason E. Pepin

Edward A. Rodriguez

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Plan for today

8.15-9.15: Jigsaw peer learning activity

9.15-9.25: Break

9.25-9.35: Activity summary

9.35-10.00 Model fit statistics demo

[And class 4 and 8 material](#)
(mass transfer, PDEs, and Fick's law)

Jigsaw activity

Check out `jigsaw.md` in the activity subdirectory for today's class

Plan

1. Count off to five in groups of two: 1, 1, 2, 2, 3, 3, 4, 4, 5, 5, 1, 1, 2, 2, ...
2. 10-20 minutes learning (use any resources, even AI)
3. Even numbered groups teach nearest odd groups for 4 minutes
4. Reverse: teaching groups learn, learning groups teach for 4 minutes
5. Quickly find the next closest group and repeat teach (4 min.) and learn (4 min.)
6. Repeat 5 until you have exchanged with all four other groups.

Diffusion exercise has been updated so code in `diff_mods.diff1D()` should be easier to understand now. See files under class 18.